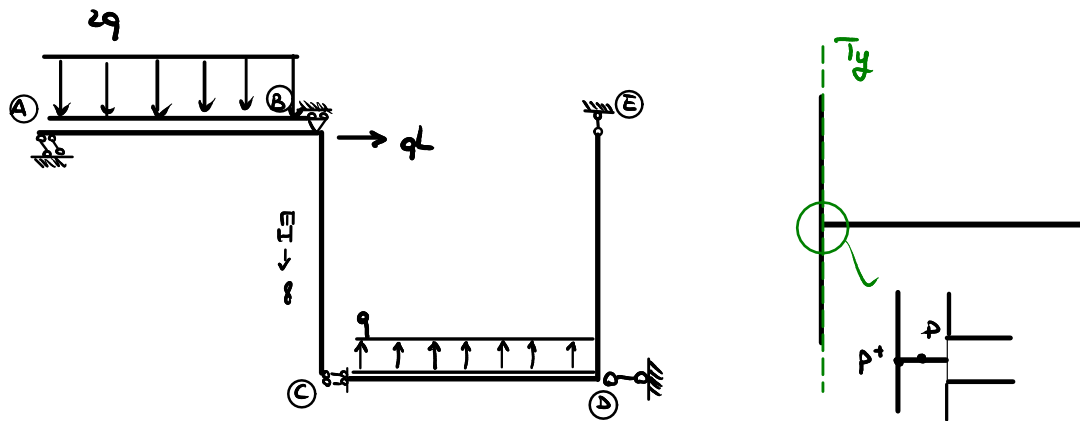
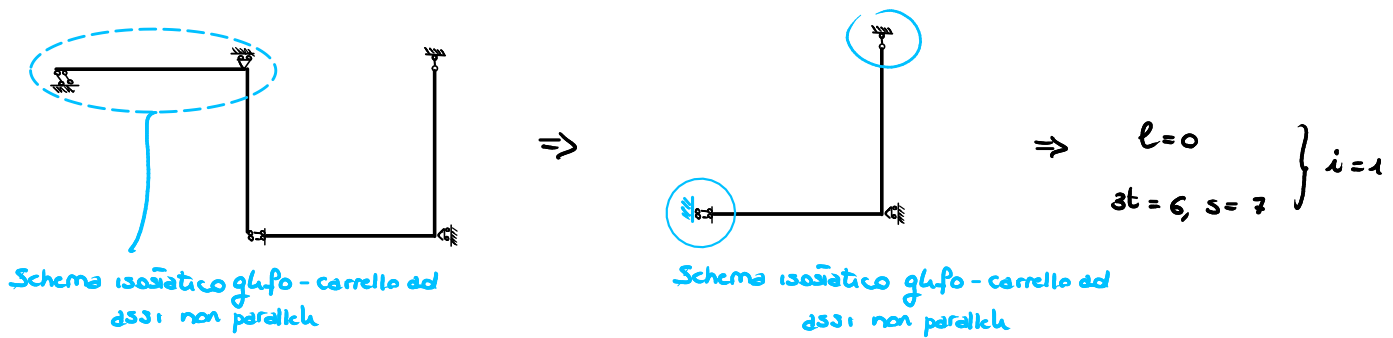


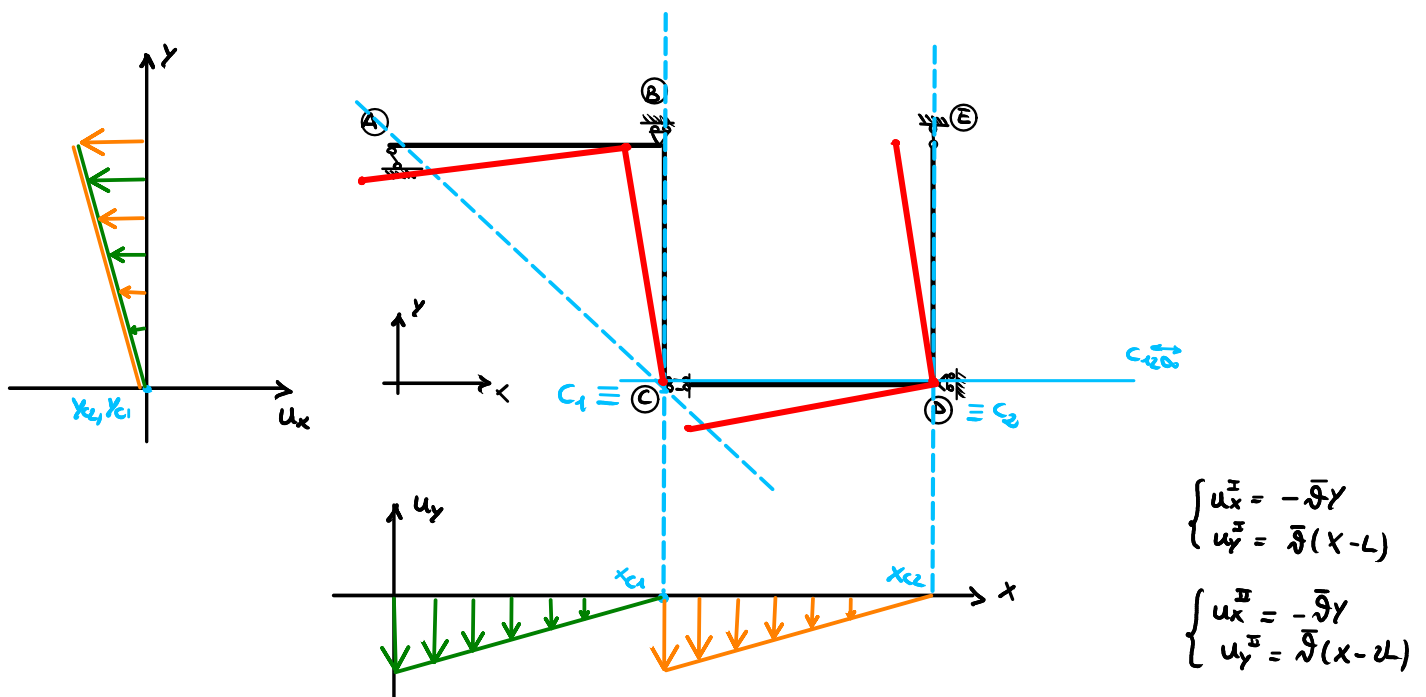
Traccia A



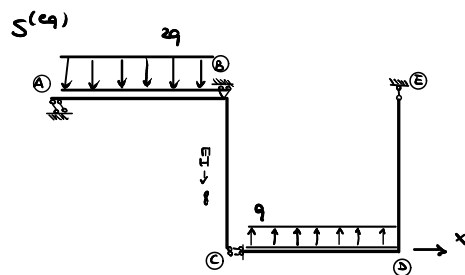
Grado di iperstaticità



Catene cinematiche

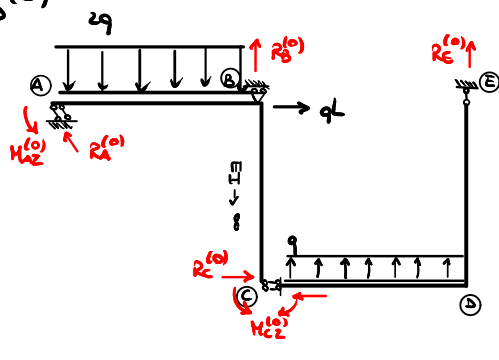


Soluzione iperstatica (metodo delle forze)

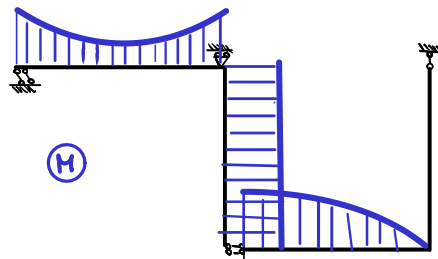
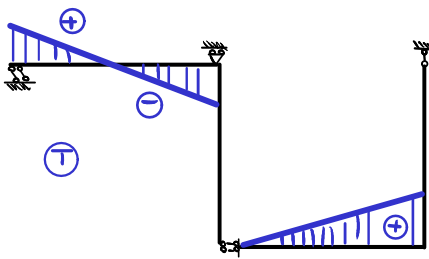
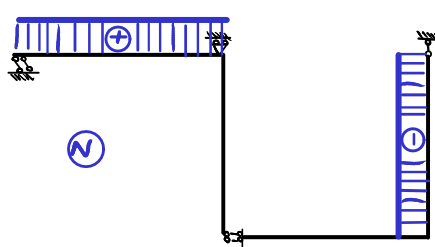
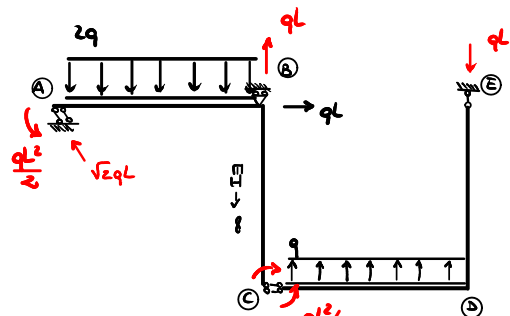


$$U \quad w_D = 0 \quad (\text{oppure } v_D = 0)$$

$S^{(0)}$



$$\begin{aligned} -\frac{RA^{(0)}l}{2} + qL + RB^{(0)} &= 0 \\ RA^{(0)}\frac{l}{2} + RB^{(0)} - 2qL &= 0 \\ M_{zC} = M_{Az}^{(0)} + 2qL \cdot \frac{l}{2} - qL^2 + M_{Cz}^{(0)} &= 0 \\ -RC^{(0)} &= 0 \\ qL + RC^{(0)} &= 0 \\ M_{zD} = -M_{Cz}^{(0)} - \frac{qL^2}{2} &= 0 \end{aligned}$$

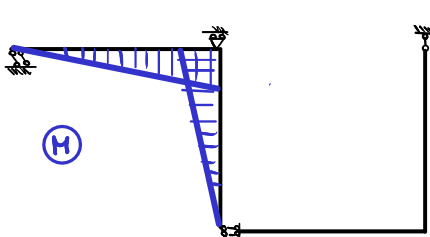
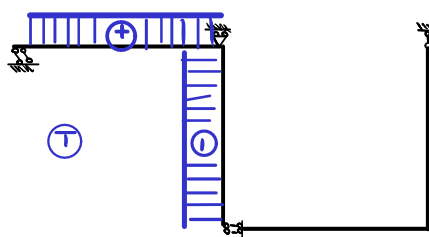
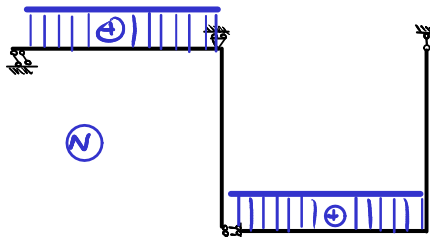
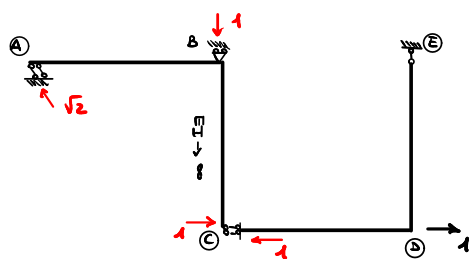
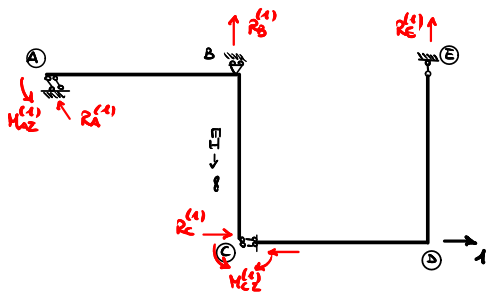


$$\begin{aligned} N_{AB}^{(0)} &= qL & N_{CB}^{(0)} &= 0 \\ N_{BC}^{(0)} &= 0 & N_{DE}^{(0)} &= -qL \end{aligned}$$

$$\begin{aligned} T_{AB}^{(0)} &= qL - 2qL & T_{CB}^{(0)} &= qL \\ T_{BC}^{(0)} &= 0 & T_{DE}^{(0)} &= 0 \end{aligned}$$

$$\begin{aligned} M_{AB}^{(0)} &= -\frac{qL^2}{2} + qL^2 - 2q\frac{L^2}{2} & M_{CB}^{(0)} &= -\frac{qL^2}{2} + \frac{qL^2}{2} \\ M_{BC}^{(0)} &= -\frac{qL^2}{2} & M_{DE}^{(0)} &= 0 \end{aligned}$$

$S^{(1)}$



$$\begin{aligned} N_{AB}^{(1)} &= 1 & N_{CB}^{(1)} &= 1 \\ N_{BC}^{(1)} &= 0 & N_{DE}^{(1)} &= 0 \end{aligned}$$

$$\begin{aligned} T_{AB}^{(1)} &= 1 & T_{CB}^{(1)} &= 0 \\ T_{BC}^{(1)} &= -1 & T_{DE}^{(1)} &= 0 \end{aligned}$$

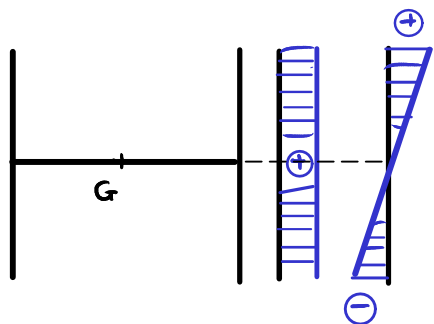
$$\begin{aligned} M_{AB}^{(1)} &= 2 & M_{CB}^{(1)} &= 0 \\ M_{BC}^{(1)} &= (L-2) & M_{DE}^{(1)} &= 0 \end{aligned}$$

Teorema dei lavori virtuali

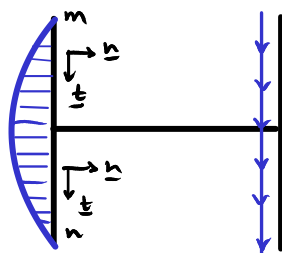
$$\delta W_{\text{ext}} = 0$$

$$\delta W_{\text{int}} = \int_{AB} \frac{M_{AB}^{(1)} (M_{AB}^{(0)} + x M_{AB}^{(1)})}{EI} dz = \int_0^L z \left(-\frac{q^2 z^2}{2} + qtz - 2q\frac{z^2}{2} + xz \right) dz \quad \left. \vphantom{\int_0^L} \right\} \delta W_{\text{ext}} = \delta W_{\text{int}} \Leftrightarrow x = \frac{qL}{2}$$

Sezione A $N_A = N_A^{(0)} + x N_A^{(1)} = \frac{3}{2}qL$, $T_A = T_A^{(0)} + x T_A^{(1)} = \frac{3}{2}qL$, $M_A = M_A^{(0)} + x M_A^{(1)} = -\frac{qL^2}{2}$

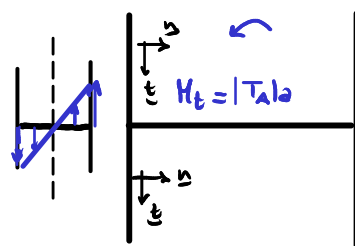


$$\sigma_z(A) = \frac{N_A}{A} + \frac{M_A}{I_x} y$$



$$\tau_{zt}^{(T)}(A) = -\frac{T_A}{SI_x} S_x^*$$

$$S_{xmn}^* = -st \left(d - \frac{t}{2} \right)$$



$$\tau_{zt}^{(0)}(A) = -2G\Theta$$

$$G\Theta = \frac{M_t}{I_t} = \frac{|T_A|d}{I_t}$$

$$I_t = \frac{1}{3} s^3 d = 2ds^3$$

Verifica

$$\sigma_z(P) = N_A/A$$

$$\textcircled{P} \quad \tau_{zt}^{(T)}(P) = -\frac{T_A}{SI_x} S_x^*(t=d) = \tau_{zy}^{(T)}$$

$$\tau_{zt}^{(0)}(P) = 0$$

$$\sigma_{eq}(P) = \sqrt{\sigma_z^2(P) + 3(\tau_{zy}^{(T)})^2} < \frac{\sigma_{sn}}{f_s}$$

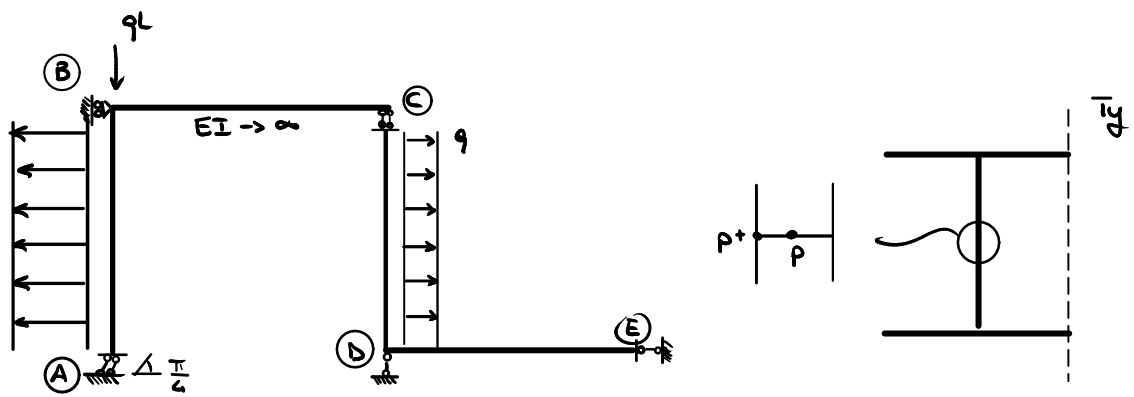
$$\sigma_z(P^*) = N_A/A$$

$$\textcircled{P^*} \quad \tau_{zt}^{(T)}(P^*) = -\frac{T_A}{SI_x} S_x^*(t=d) = \tau_{zy}^{(T)}$$

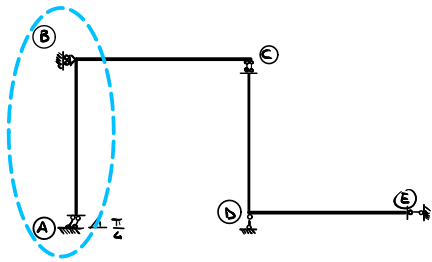
$$\tau_{zt}^{(0)}(P^*) = -2G\Theta \left(-\frac{s}{2} \right) = \tau_{zy}^{(0)}$$

$$\sigma_{eq}(P) = \sqrt{\sigma_z^2(P) + 3(\tau_{zy}^{(T)} + \tau_{zy}^{(0)})^2} < \frac{\sigma_{sn}}{f_s}$$

Traccia B

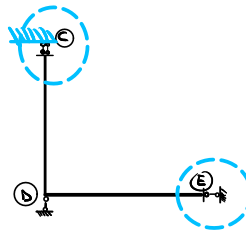


Grado di iperstaticità



Schema ipostatico glifo-carrello ad assi non paralleli

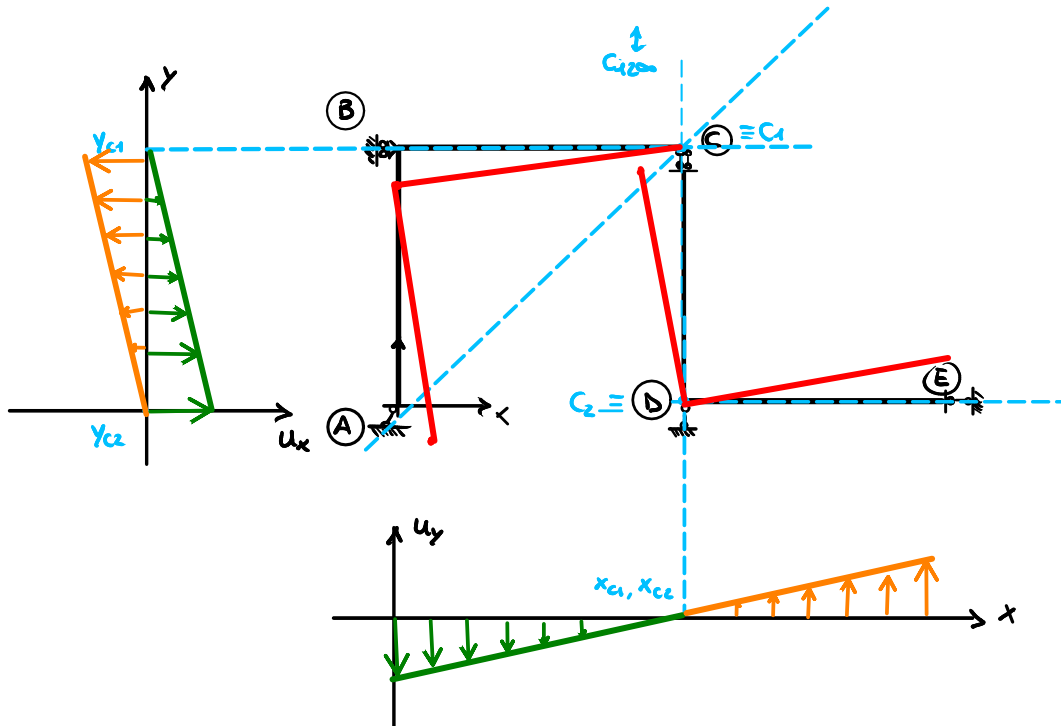
$\Rightarrow$



Schema ipostatico glifo-carrello ad assi non paralleli

$$\Rightarrow \left. \begin{array}{l} l=0 \\ 3t=6, s=7 \end{array} \right\} i=1$$

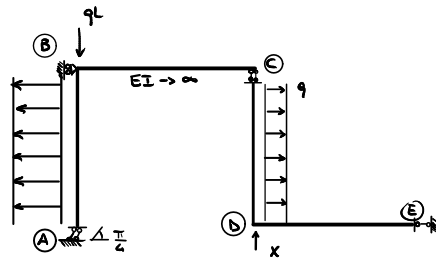
Catene cinematiche



$$\begin{cases} u_x^I = \bar{q}(L-y) \\ u_y^I = -\bar{q}(L-x) \end{cases}$$

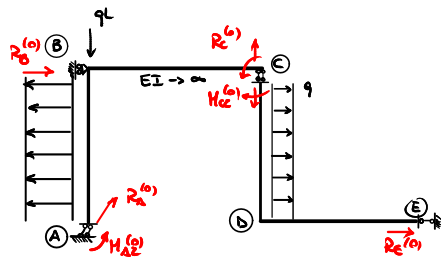
$$\begin{cases} u_x^{II} = -\bar{q}y \\ u_y^{II} = \bar{q}(x-L) \end{cases}$$

Soluzione iperstatica (metodo delle forze)

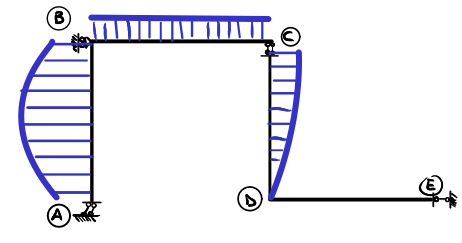
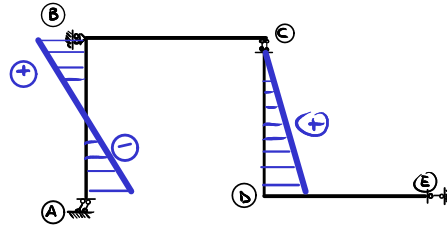
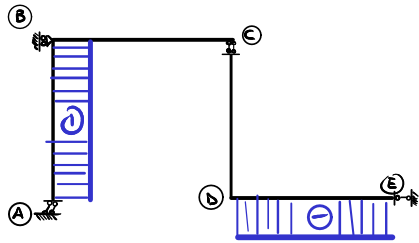
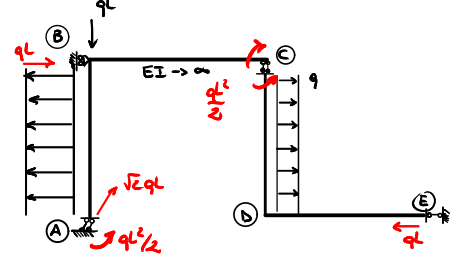


$$U \quad w_D = 0 \\ (\text{oppure } v_D = 0)$$

$S^{(0)}$



$$\begin{aligned} R_A^{(0)} \frac{\sqrt{2}}{2} - 2qL + R_C^{(0)} &= 0 \\ R_A^{(0)} \frac{\sqrt{2}}{2} - qL + R_C^{(0)} &= 0 \\ H_{AE}^{(0)} - 2q \cdot \frac{L}{2} + qL^2 + H_{CE}^{(0)} &= 0 \\ qL + R_C^{(0)} &= 0 \\ -R_C^{(0)} &= 0 \\ H_{AE}^{(0)} - H_{CE}^{(0)} - \frac{qL^2}{2} &= 0 \end{aligned}$$

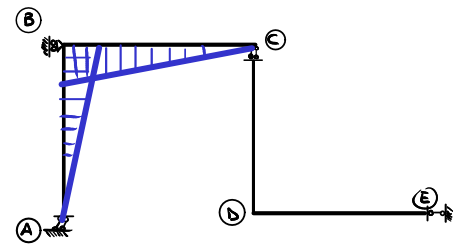
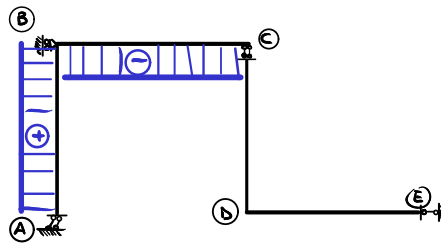
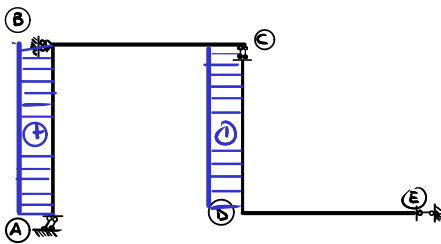
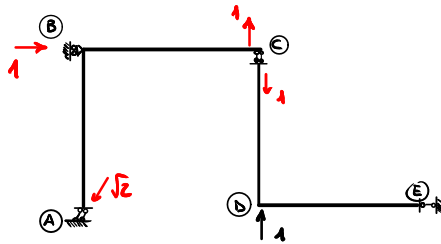


$$\begin{aligned} N_{AB}^{(0)} &= -qL & N_{CB}^{(0)} &= 0 \\ N_{BC}^{(0)} &= 0 & N_{CE}^{(0)} &= -qL \end{aligned}$$

$$\begin{aligned} T_{AB}^{(0)} &= -qL + 2qz & T_{CB}^{(0)} &= qz \\ T_{BC}^{(0)} &= 0 & T_{CE}^{(0)} &= 0 \end{aligned}$$

$$\begin{aligned} M_{AB}^{(0)} &= -\frac{qL^2}{2} - qLz + \frac{2qz^2}{2} & M_{CB}^{(0)} &= -\frac{qL^2}{2} + qLz \\ M_{BC}^{(0)} &= -\frac{qL^2}{2} & M_{CE}^{(0)} &= 0 \end{aligned}$$

$S^{(1)}$



$$\begin{aligned} N_{AB}^{(1)} &= 1 & N_{CB}^{(1)} &= -1 \\ N_{BC}^{(1)} &= 0 & N_{CE}^{(1)} &= 0 \end{aligned}$$

$$\begin{aligned} T_{AB}^{(1)} &= 1 & T_{CB}^{(1)} &= 0 \\ T_{BC}^{(1)} &= -1 & T_{CE}^{(1)} &= 0 \end{aligned}$$

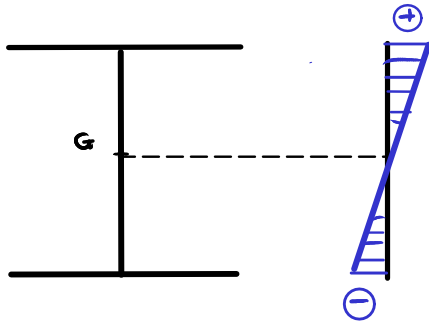
$$\begin{aligned} M_{AB}^{(1)} &= z & M_{CB}^{(1)} &= 0 \\ M_{BC}^{(1)} &= L-z & M_{CE}^{(1)} &= 0 \end{aligned}$$

Teorema dei lavori virtuali

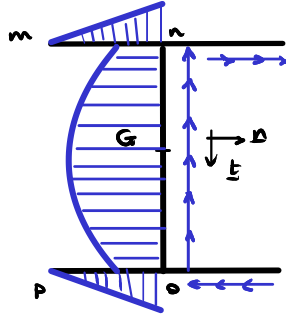
$$\delta W_{ve} = 0$$

$$\delta W_{vi} = \int_{AB} \frac{M_{AB}^{(1)} (M_{AB}^{(0)} + x M_{AB}^{(1)})}{EI} dz = \int_0^L \frac{z \left(-\frac{qL^2}{2} - qLz + \frac{2qz^2}{2} + xz \right)}{EI} dz \quad \left. \vphantom{\int_0^L} \right\} \delta W_{ve} = \delta W_{vi} \Leftrightarrow x = qL$$

Sezione C- $N = N_C^{(0)} + x N_C^{(1)} = 0$, $T_C = T_C^{(0)} + x T_C^{(1)} = -qL$, $M_C = M_C^{(0)} + x M_C^{(1)} = -\frac{qL^2}{2}$

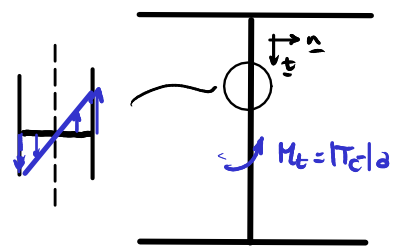


$$\sigma_z(C) = \frac{N_C}{A} + \frac{M_C}{I_x} y$$



$$\tau_{zt}^{(T)}(C) = -\frac{T_C}{SI_x} S_x^*$$

$$\begin{aligned} S_{xmn}^* &= -st a \\ S_{xno}^* &= -2sa^2 - st \left(a - \frac{t}{2} \right) \\ S_{xop}^* &= -sa^2 + sta \end{aligned}$$



$$\tau_{zt}^{(0)}(A) = -2G\Theta n$$

$$G\Theta = \frac{M_t}{I_t} = \frac{|T_C| a}{I_t}$$

$$I_t = \frac{1}{3} s^3 a = 2as^3$$

Verifica

$$\sigma_z(P) = 0$$

$$\begin{aligned} \textcircled{P} \quad \tau_{zt}^{(T)}(P) &= -\frac{T_C}{SI_x} S_{xno}^*(t=a) = \tau_{zy}^{(T)} \\ \tau_{zt}^{(0)}(P) &= 0 \end{aligned}$$

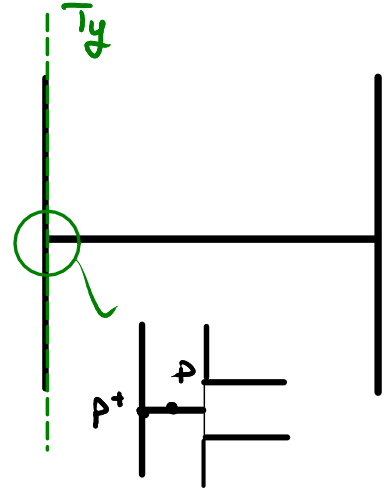
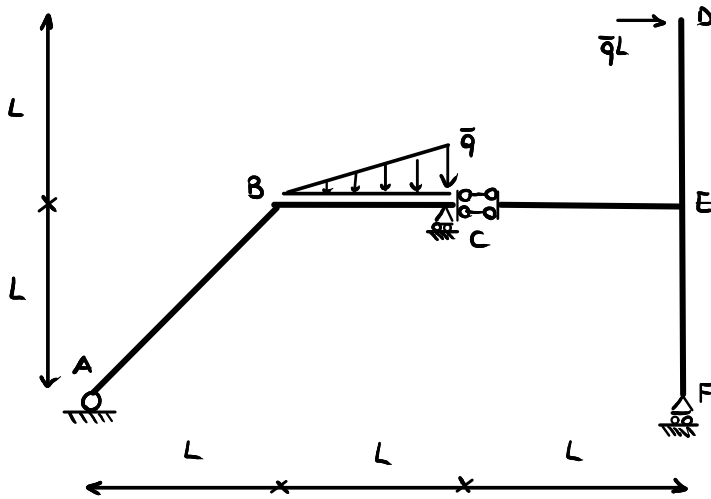
$$\sigma_{eq}(P) = \sqrt{4 (\tau_{zy}^{(T)})^2} = 2\tau_{zy}^{(T)} < \frac{\sigma_{sn}}{f_s}$$

$$\sigma_z(P^*) = 0$$

$$\begin{aligned} \textcircled{P^*} \quad \tau_{zt}^{(T)}(P^*) &= -\frac{T_C}{SI_x} S_{xno}^*(t=a) = \tau_{zy}^{(T)} \\ \tau_{zt}^{(0)}(P^*) &= -2G\Theta \left(-\frac{s}{2} \right) = \tau_{zy}^{(0)} \end{aligned}$$

$$\sigma_{eq}(P) = \sqrt{4 (\tau_{zy}^{(T)} + \tau_{zy}^{(0)})^2} < \frac{\sigma_{sn}}{f_s}$$

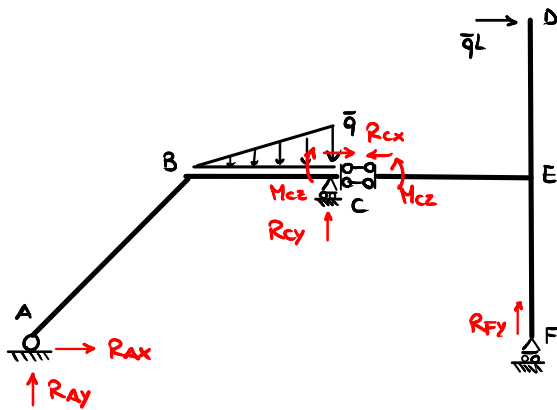
Traccia C



IPERSTATICITA'

$i = 0$, isostatica

REAZIONI VINCOLARI



$$R_{ax} + R_{cx} = 0$$

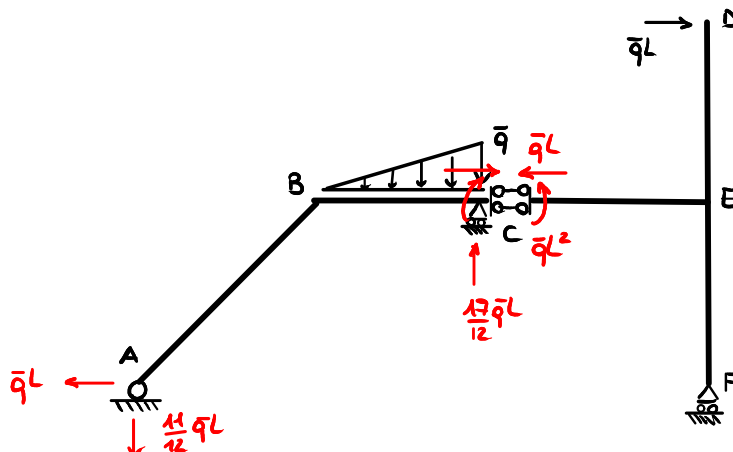
$$R_{ay} - \frac{\bar{q}L}{2} + R_{cy} = 0$$

$$M_z(A) = -\frac{\bar{q}L}{2} \left(L + \frac{2L}{3} \right) + R_{cy} 2L - M_{cz} - R_{cx}L = 0$$

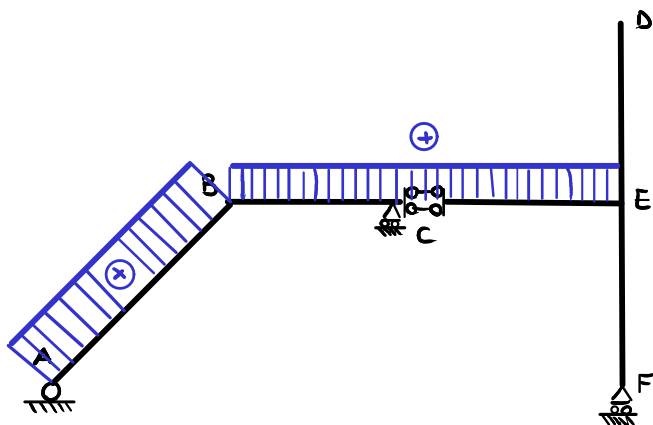
$$-R_{cx} + \bar{q}L = 0$$

$$R_{fy} = 0$$

$$M_z(E) = -\bar{q}L^2 + M_{cz} = 0$$



CARATTERISTICHE DELLA SOLLECITAZIONE



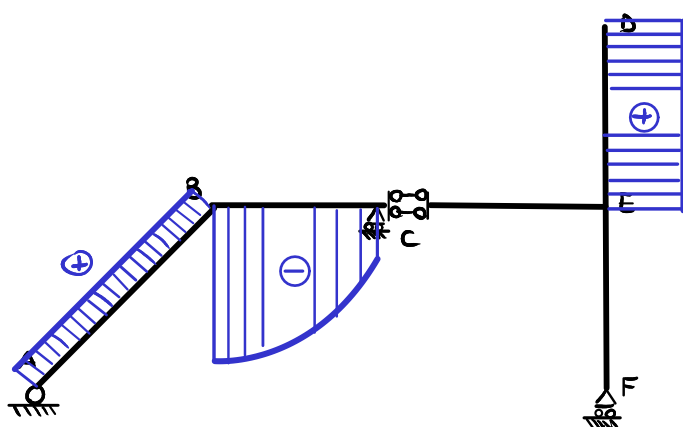
$$N_{AB} = \frac{\bar{q}L}{\sqrt{2}} + \frac{11}{12\sqrt{2}} \bar{q}L = \frac{23}{12\sqrt{2}} \bar{q}L$$

$$N_{BC} = \bar{q}L$$

$$N_{CE} = \bar{q}L$$

$$N_{DE} = 0$$

$$N_{EF} = 0$$



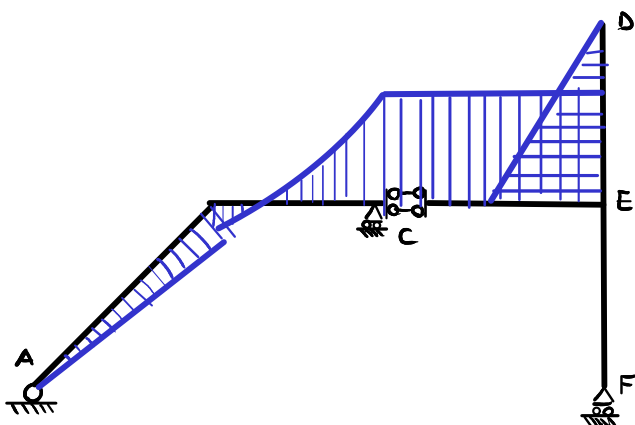
$$T_{AB} = \frac{\bar{q}L}{\sqrt{2}} - \frac{11}{12\sqrt{2}} \bar{q}L = \frac{\bar{q}L}{12\sqrt{2}}$$

$$T_{BC} = -\frac{11}{12} \bar{q}L - \frac{q(z)z}{2} = -\frac{11}{12} \bar{q}L - \frac{\bar{q}z^2}{2L}$$

$$T_{CE} = 0$$

$$T_{DE} = \bar{q}L$$

$$T_{EF} = 0$$



$$M_{AB} = \frac{\bar{q}L}{\sqrt{2}} z - \frac{11}{12\sqrt{2}} \bar{q}L z = \frac{\bar{q}L}{12\sqrt{2}} z$$

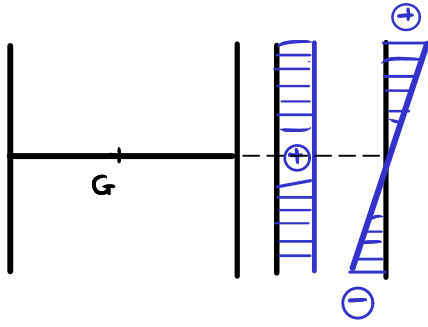
$$M_{BC} = \bar{q}L^2 - \frac{11}{12} \bar{q}L(L+z) - \frac{q(z)z}{2} \frac{z}{3} = \frac{\bar{q}L^2}{12} - \frac{11}{12} \bar{q}Lz - \frac{\bar{q}z^3}{6L}$$

$$M_{CE} = -\bar{q}L^2$$

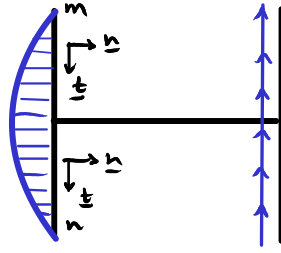
$$M_{DE} = \bar{q}Lz$$

$$M_{EF} = 0$$

Sezione C- $N = \bar{q}L$, $T_C = -\frac{17}{12} \bar{q}L$, $M_C = -\bar{q}L^2$

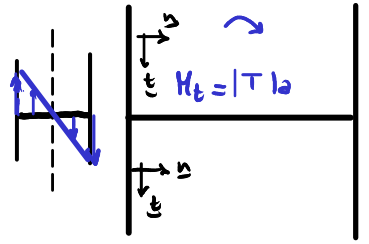


$$\sigma_z(A) = \frac{N_c^-}{A} + \frac{Y_c^-}{I_x} y$$



$$\tau_{zt}^{(T)}(C^-) = - \frac{T}{S I_x} S_x^*$$

$$S_{xmn}^* = -st(a - \frac{t}{2})$$

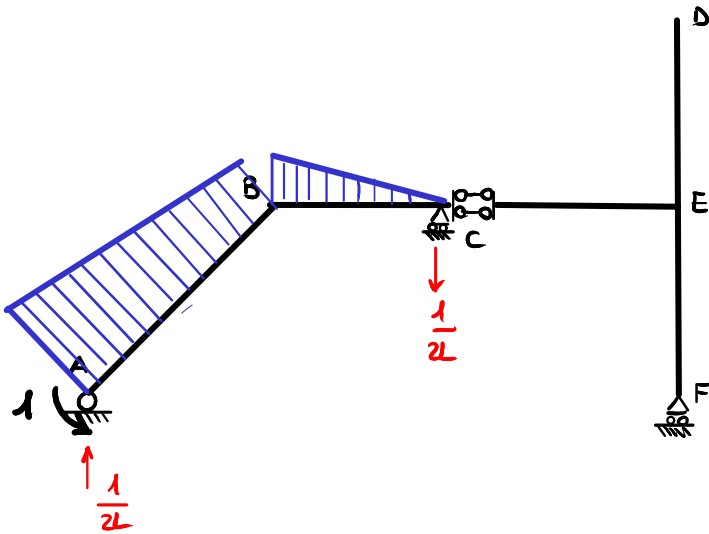


$$\tau_{\text{gt}}^{(\Theta)}(\cdot) = -2G\Theta n$$

$$G\Theta = \frac{M_t}{I_t} = \frac{1-a}{I_t}$$

$$I_t = \frac{1}{3} s^3 \phi d = 2as^3$$

CALCOLO DELLA ROTAZIONE Φ_A

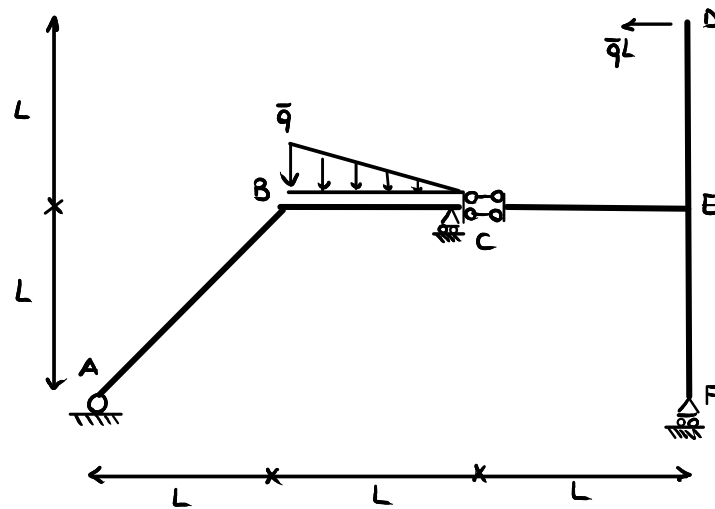


$$M_{AB} = -1 + \frac{1}{2\sqrt{2}L} z$$

$$M_{BC} = - \frac{L-z}{2L}$$

$$\Phi_A = \frac{1}{EI} \int_{AB} M_{AB} H_{AB}^{(v)} dz$$

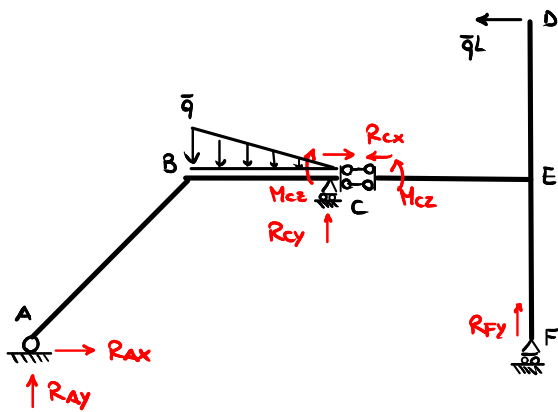
Traccia D



IPERSTATICITA'

$i = 0$, ipostatica

REAZIONI VINCOLARI



$$R_{Ax} + R_{Cx} = 0$$

$$R_{Ay} - \frac{\bar{q}L}{2} + R_{Cy} = 0$$

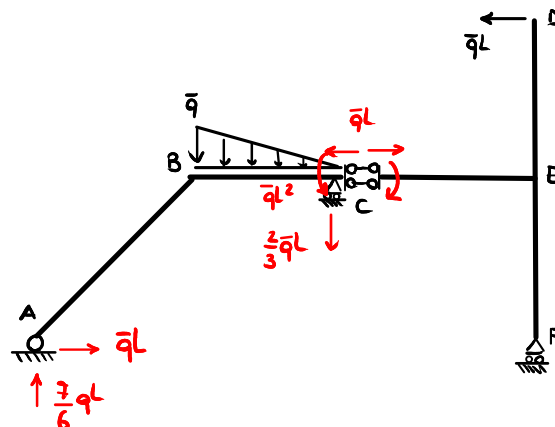
$$M_z(A) = -\frac{\bar{q}L}{2} \cdot \left(L + \frac{L}{3}\right) + R_{Cy} 2L - M_{Cx} - R_{Cx}L = 0$$

$$-R_{Cx} - \bar{q}L = 0$$

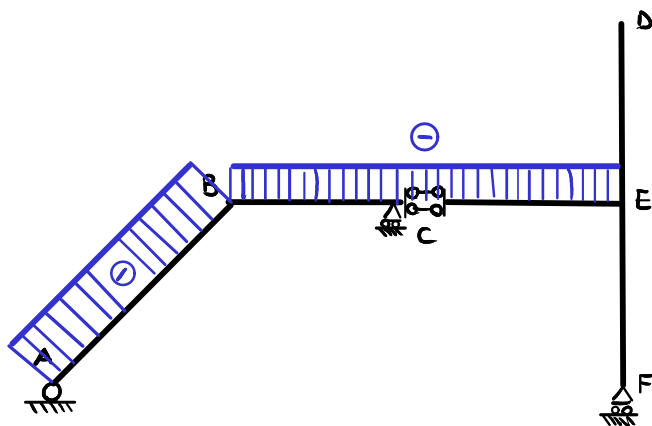
$$R_{Fy} = 0$$

$$M_z(E) = +\bar{q}L^2 + M_{Cx} = 0$$

$$q(z) = \bar{q} \frac{L-z}{L}$$



CARATTERISTICHE DELLA SOLLECITAZIONE



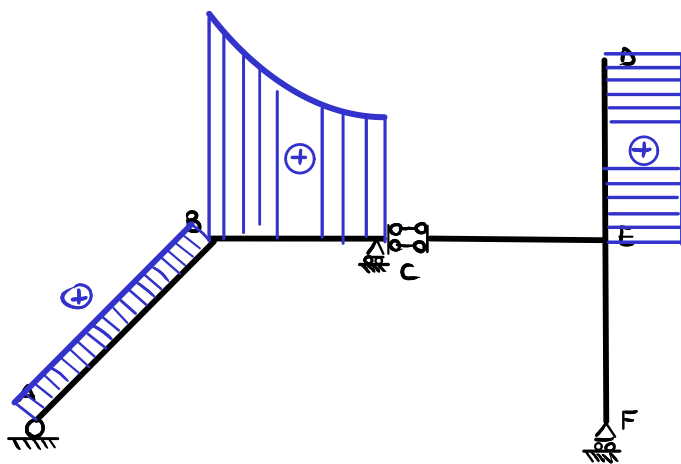
$$N_{AB} = -\frac{7}{6\sqrt{2}} \bar{q}L - \frac{\bar{q}L}{\sqrt{2}} = -\frac{13}{6\sqrt{2}} \bar{q}L$$

$$N_{BC} = -\bar{q}L$$

$$N_{CE} = -\bar{q}L$$

$$N_{DE} = 0$$

$$N_{EF} = 0$$



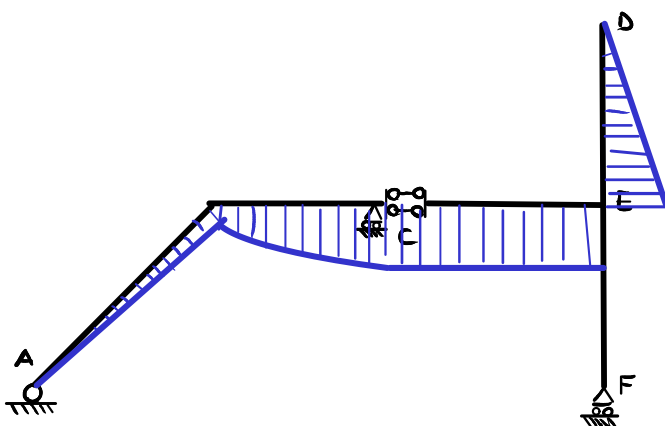
$$T_{AB} = \frac{7}{6\sqrt{2}} \bar{q}L - \frac{\bar{q}L}{\sqrt{2}} = \frac{\bar{q}L}{6\sqrt{2}}$$

$$T_{BC} = \frac{2}{3} \bar{q}L + q(z) \frac{(L-z)}{2} = \frac{2}{3} \bar{q}L + \bar{q} \frac{(L-z)^2}{2L}$$

$$T_{CE} = 0$$

$$T_{DE} = -\bar{q}L$$

$$T_{EF} = 0$$



$$M_{AB} = \frac{7}{6\sqrt{2}} \bar{q}Lz - \frac{\bar{q}Lz}{\sqrt{2}} = \frac{\bar{q}Lz}{6\sqrt{2}}$$

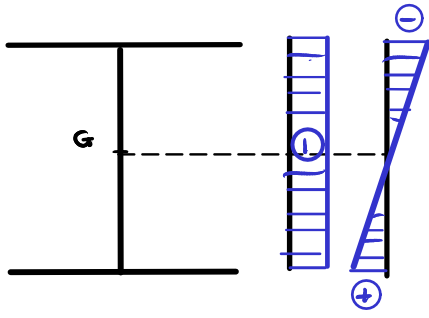
$$M_{BC} = \bar{q}L^2 - \frac{2}{3} \bar{q}L(L-z) - \frac{q(z)(L-z)}{2} \frac{(L-z)}{3} \\ = \bar{q}L^2 - \frac{2}{3} \bar{q}L(L-z) - \frac{\bar{q}(L-z)^3}{6L}$$

$$M_{CE} = \bar{q}L^2$$

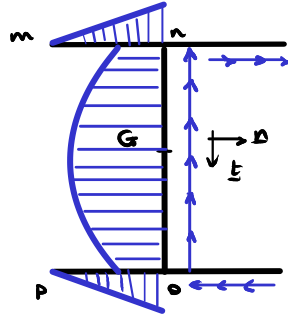
$$M_{DE} = -\bar{q}Lz$$

$$M_{EF} = 0$$

Sezione C- $N = -qL$, $T_C = \frac{2}{3} qL$, $M_C = +qL^2$

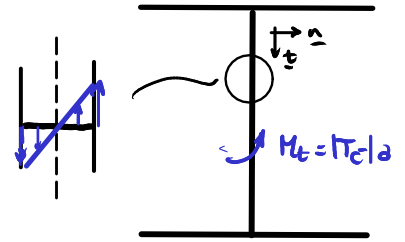


$$\sigma_z(C) = \frac{N_C}{A} + \frac{M_C}{I_x} y$$



$$\tau_{zt}^{(T)}(C) = - \frac{T_C}{SI_x} S_x^*$$

$$\begin{aligned} S_{xmn}^* &= -st a \\ S_{xno}^* &= -2sa^2 - st \left(a - \frac{t}{2}\right) \\ S_{xop}^* &= -sa^2 + sta \end{aligned}$$

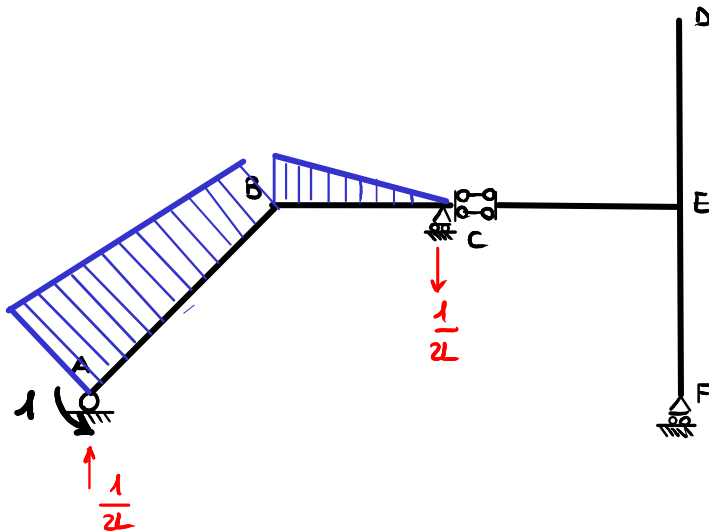


$$\tau_{zt}^{(\Theta)}(A) = -2G\Theta n$$

$$G\Theta = \frac{M_t}{I_t} = \frac{|T_C| a}{I_t}$$

$$I_t = \frac{1}{3} s^3 6a = 2as^3$$

CALCOLO DELLA ROTAZIONE Φ_A



$$M_{AB} = -1 + \frac{1}{2\sqrt{2}L} z$$

$$M_{BC} = - \frac{L-z}{2L}$$

$$\Phi_A = \frac{1}{EI} \int_{AB} M_{AB} M_{AB}^{(v)} dz$$